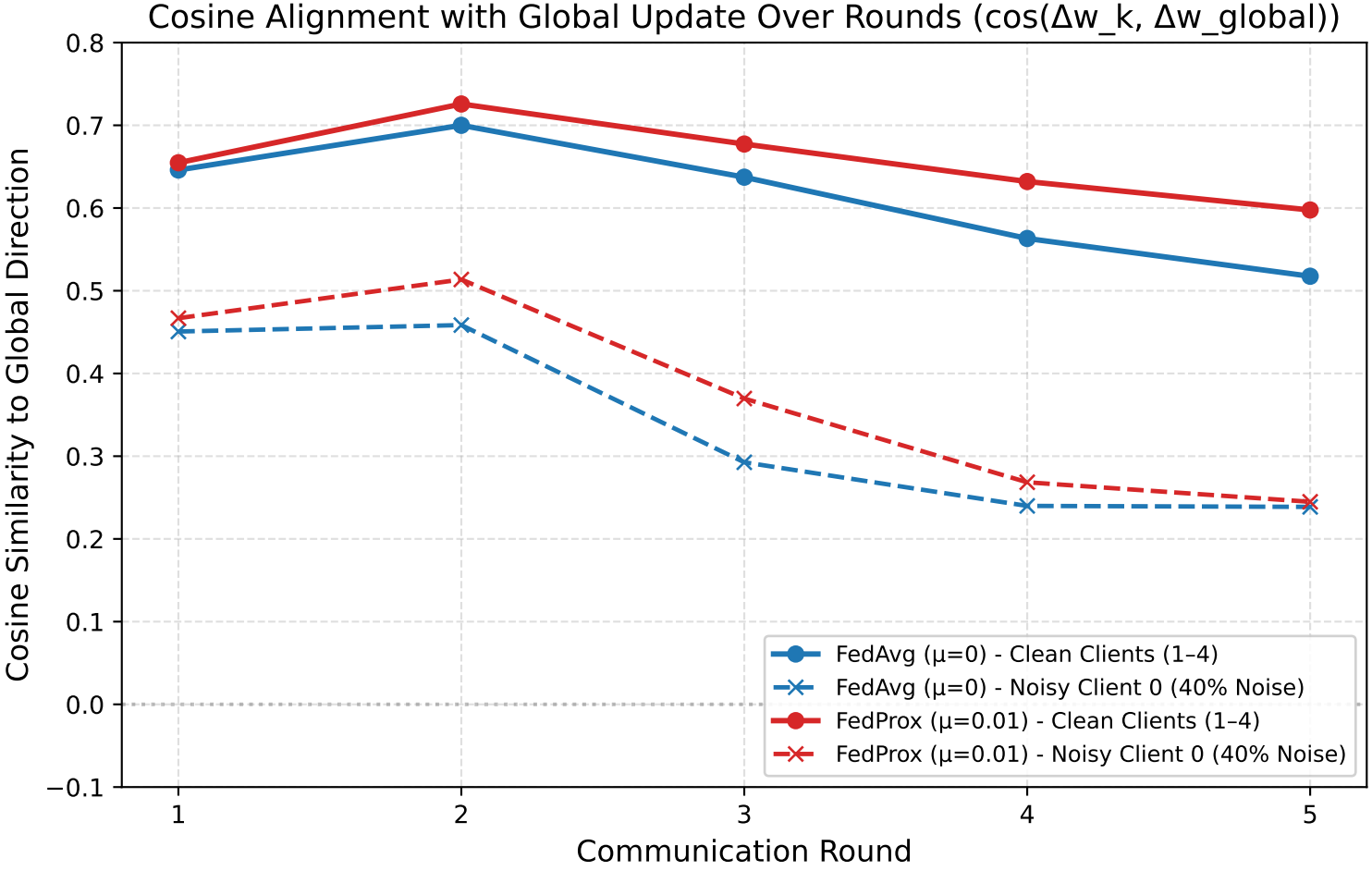
Cosine Alignment with Global Update Over Rounds (cos(Δw_k, Δw_global))
Cosine Similarity to Global Direction
Communication Round
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−0.1
1
2
3
4
5
FedAvg (μ=0) - Clean Clients (1–4)
FedAvg (μ=0) - Noisy Client 0 (40% Noise)
FedProx (μ=0.01) - Clean Clients (1–4)
FedProx (μ=0.01) - Noisy Client 0 (40% Noise)